

# HW3 Findings Draft

SID4=6812, SEED=6812, SLICE=812, HP\_ID=2, CLS\_A=2, CLS\_B=9

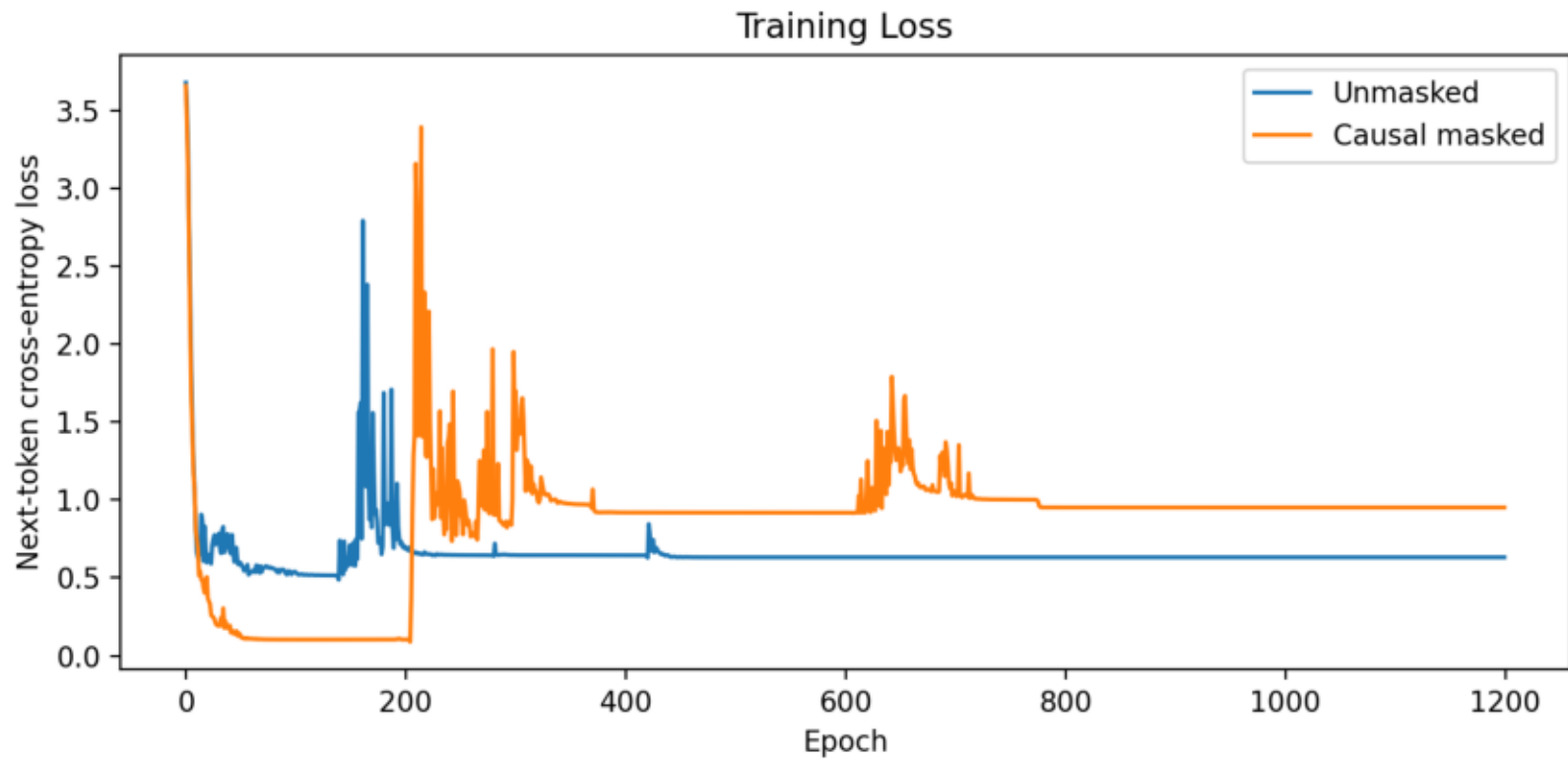
# Prompt Engineering Findings

I tried two examples for each prompting technique. Zero-shot was the quickest to write, but it gave the model the least guidance. Few-shot prompting was more controlled because the examples showed the answer format. Chain-of-thought and zero-shot CoT gave longer answers, but the extra steps made the reasoning easier to check. Meta-prompting was useful for tasks where I wanted the model to make a plan or rubric before answering. Tree of Thoughts was the most detailed because it compared multiple possible solution paths before picking an answer. The exact prompts and model outputs are saved in `prompt_engineering_outputs.csv`.

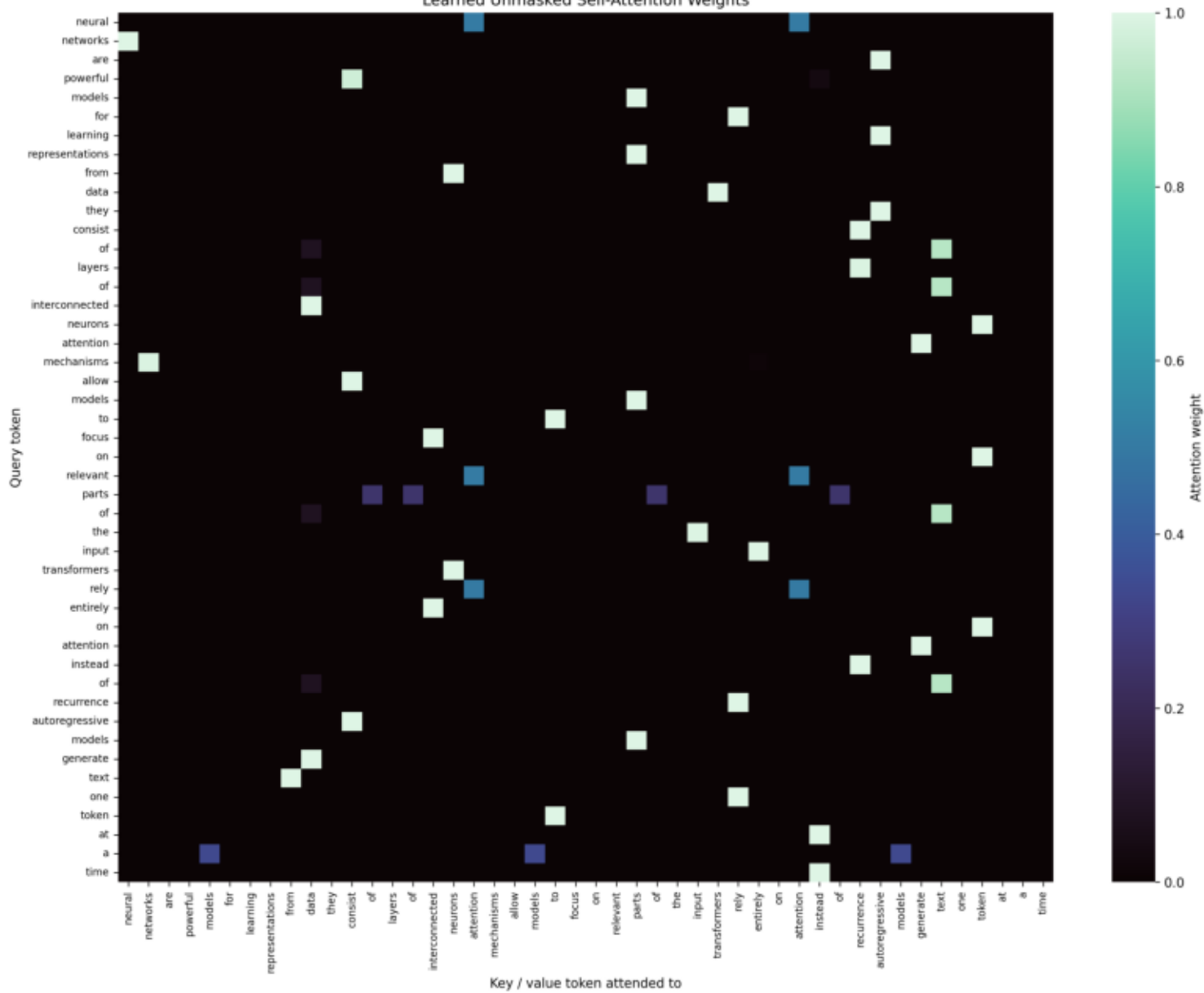
# Self-Attention Findings

After word-level tokenization, the dataset had 46 tokens and 39 unique words. I trained two small next-token models. Both used learned embeddings and learned Q, K, and V projections. The final unmasked loss was 0.6318, and the final causal masked loss was 0.9518. In the unmasked heatmap, tokens are allowed to attend anywhere in the sequence, including future positions. In the causal masked heatmap, the upper-right part of the matrix is zero because future tokens are blocked. I checked this directly: the largest attention value above the diagonal was 0.00000000.

Training Loss



Learned Unmasked Self-Attention Weights



### Causal Masked Self-Attention Heatmap

Learned Causal Masked Self-Attention Weights

